



EAST TEXAS A&M
— UNIVERSITY —

CSCI 352.61E

DIGITAL FORENSICS

System Forensics, Investigation, and Response

Lesson 8 Windows Forensics

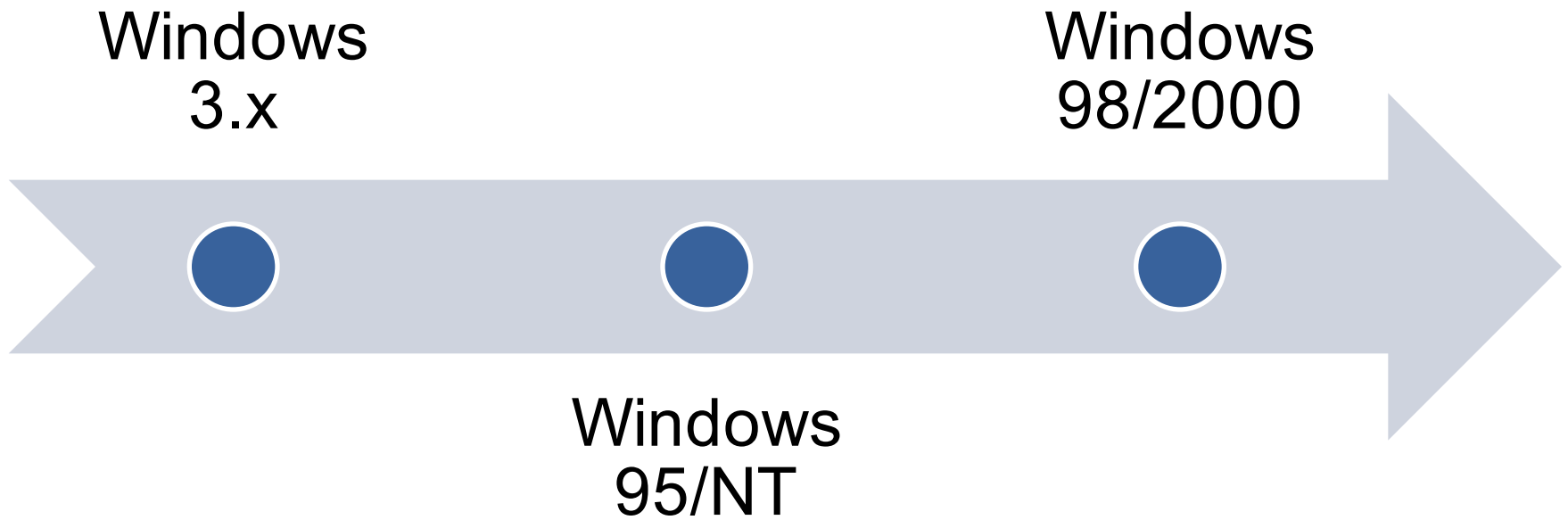
Learning Objective

- Summarize various types of digital forensics.

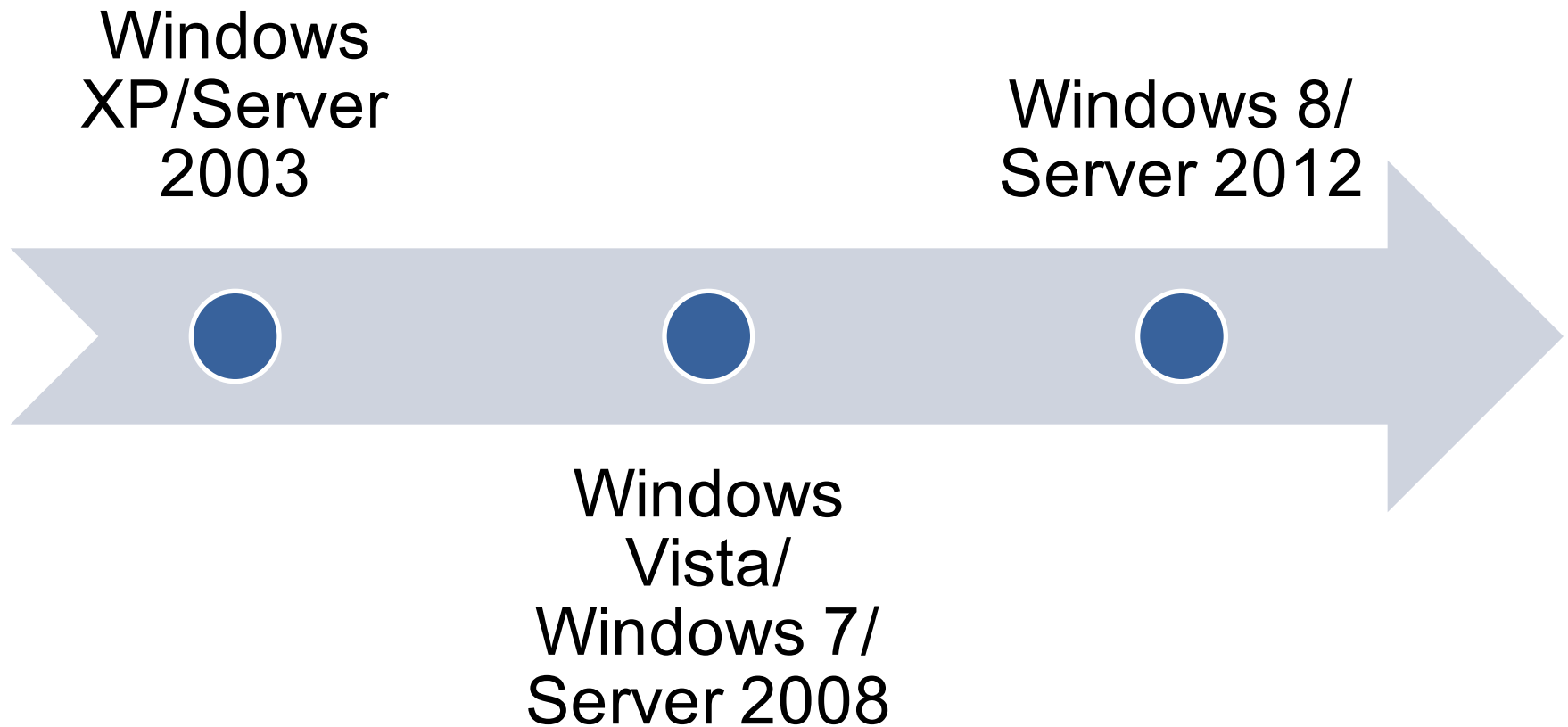
Key Concepts

- Windows details
- Evidence in volatile data
- Windows swap file
- Windows logs and directories
- Windows Registry

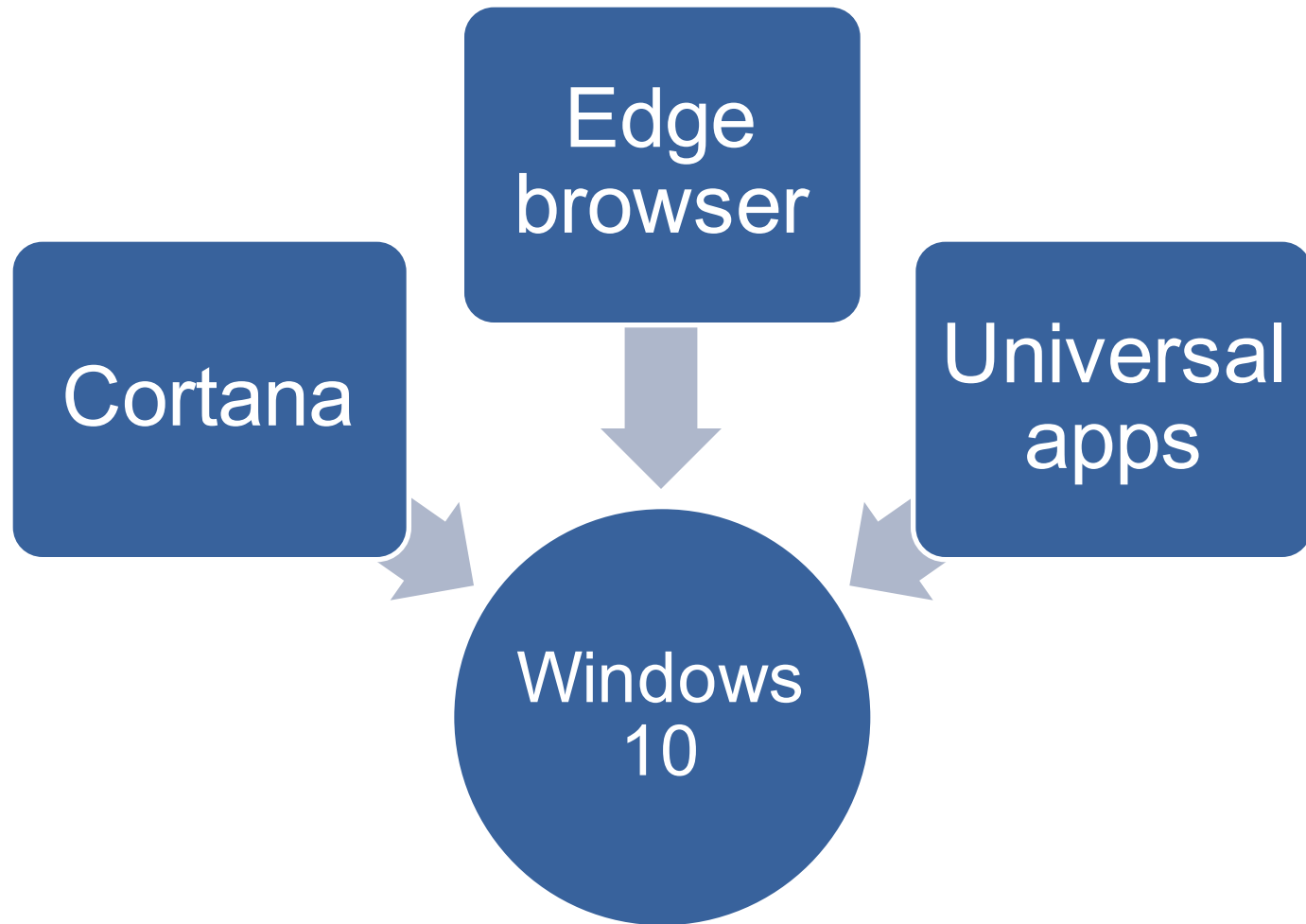
History of Windows



History of Windows (Cont.)



History of Windows (Cont.)



Issues Pertinent to Forensics

- Does the Windows version in question support 64-bit processing?
- Does it have a firewall? If so, is the firewall automatically on?
- Does the version of Windows support the Encrypted File System (EFS)?

Windows Details

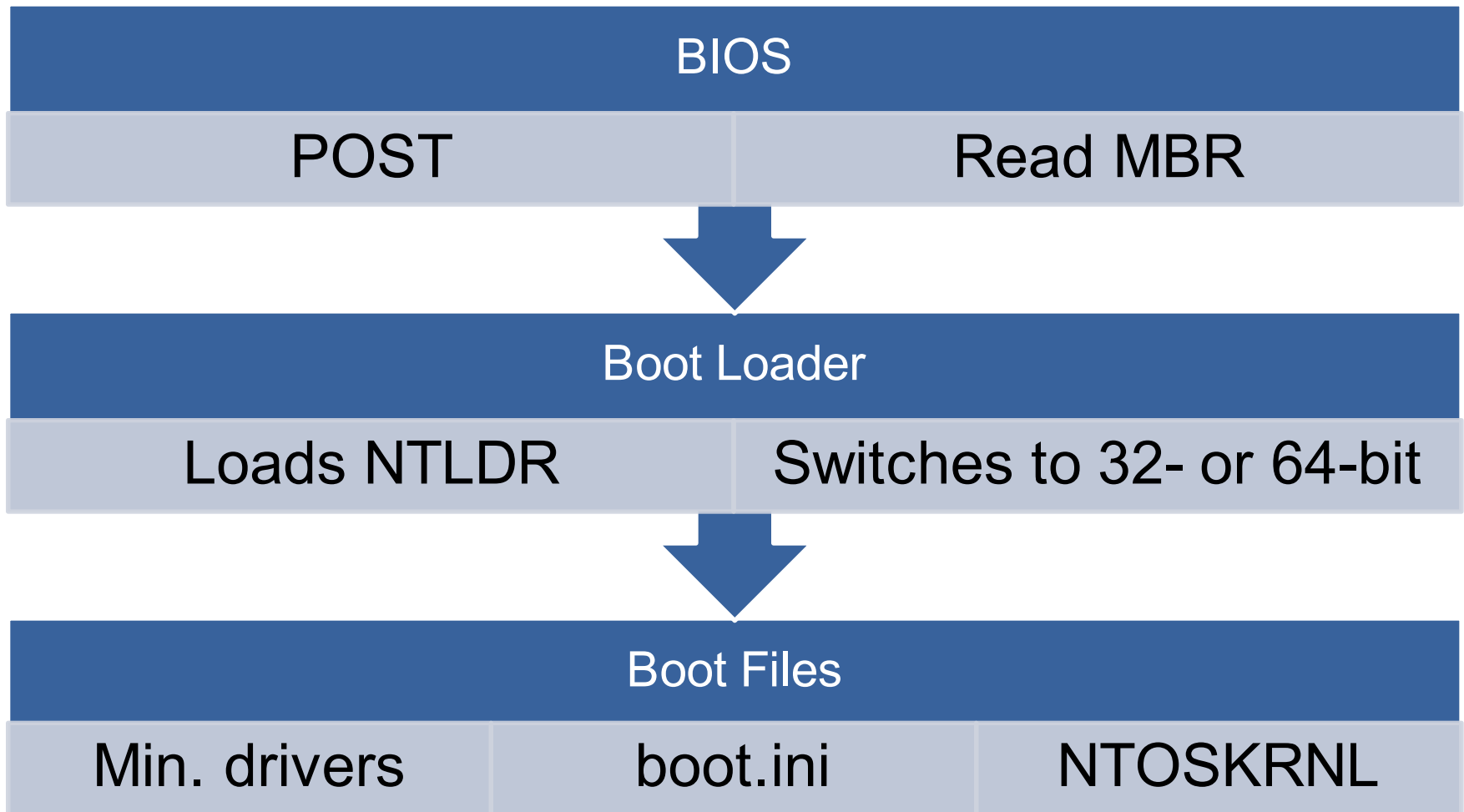
32-bit

- Addresses up to 4,294,967,295 bytes
- Limited to 4 gigabytes (GB) of RAM
- Referred to as x86

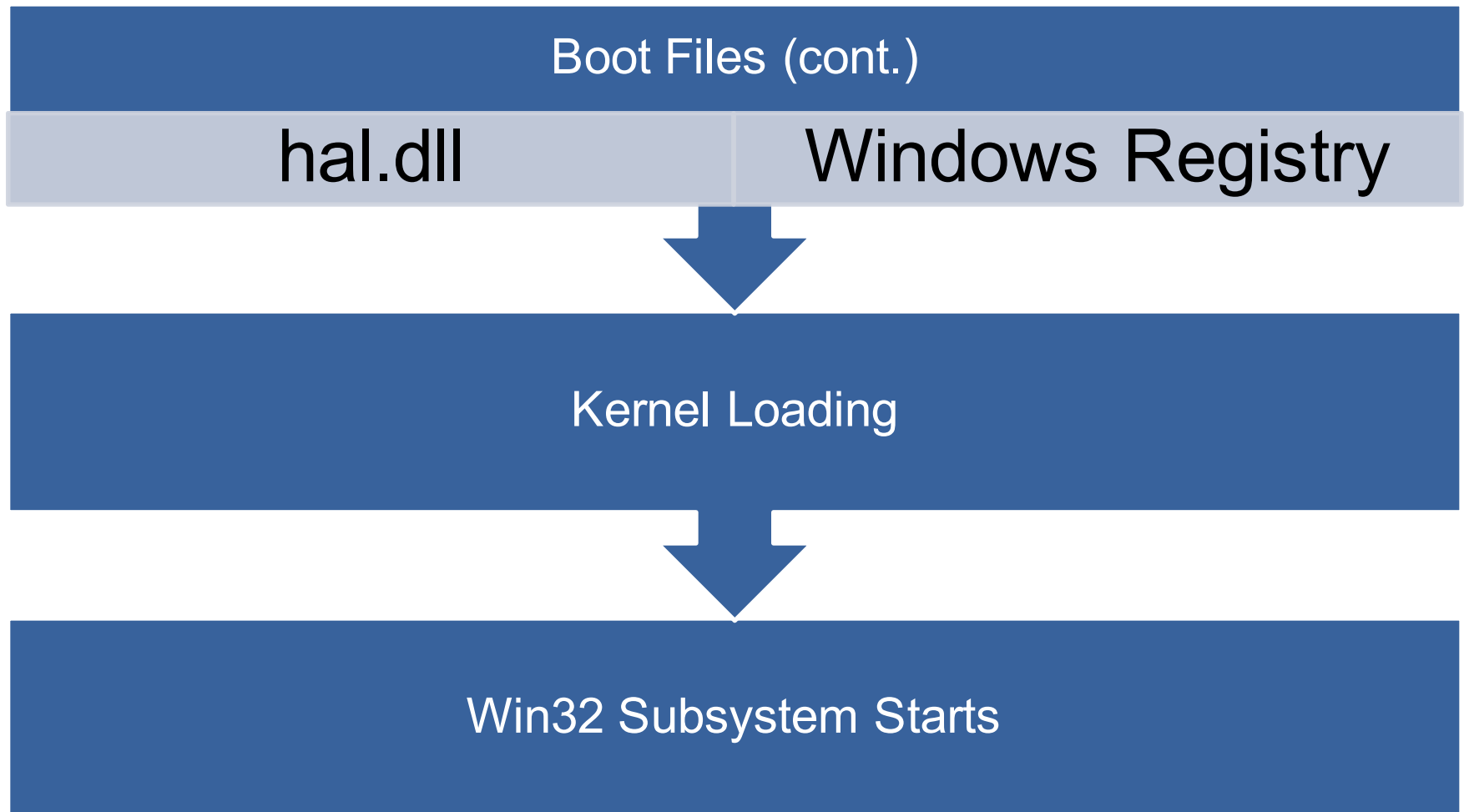
64-bit

- Addresses up to 18,446,744,073,709,551,616 bytes
- Referred to as x64

Windows Boot Process



Windows Boot Process (Cont.)



Important Files

Ntdetect.com

Ntbootdd.sys

Ntoskrnl.exe

Hal.dll

Smss.exe

Winlogon.exe

Lsass.exe

Explorer.exe

Crss.exe

Volatile Memory

- Live system forensic technique in which you:
 - Collect a memory dump
 - Compute the hash
 - Perform analysis in an isolated environment

Offline Vs. Volatile Data Analysis

Offline Data Analysis	Volatile Data Analysis
Nonvolatile data	Volatile data
Isolation is relatively easy for experienced specialist	Isolation is difficult
Repeatable	Not repeatable
Uses no software that existed on the system during the time frame of investigation	Uses software that existed on the system during the time frame of the investigation

Volatile Memory (cont.)

Stack (S)

- Allocated based on last-in, first-out (LIFO)
- Most dynamic area of memory process

Heap (H)

- Data can exist between function calls
- Is less stable than data in the data segment

Live Forensics Tools

- PsList - processes
- PsInfo – operating system details
- ListDLLs – loaded DLLs
- PsLoggedOn – login information
- netstat – network connections

PsList

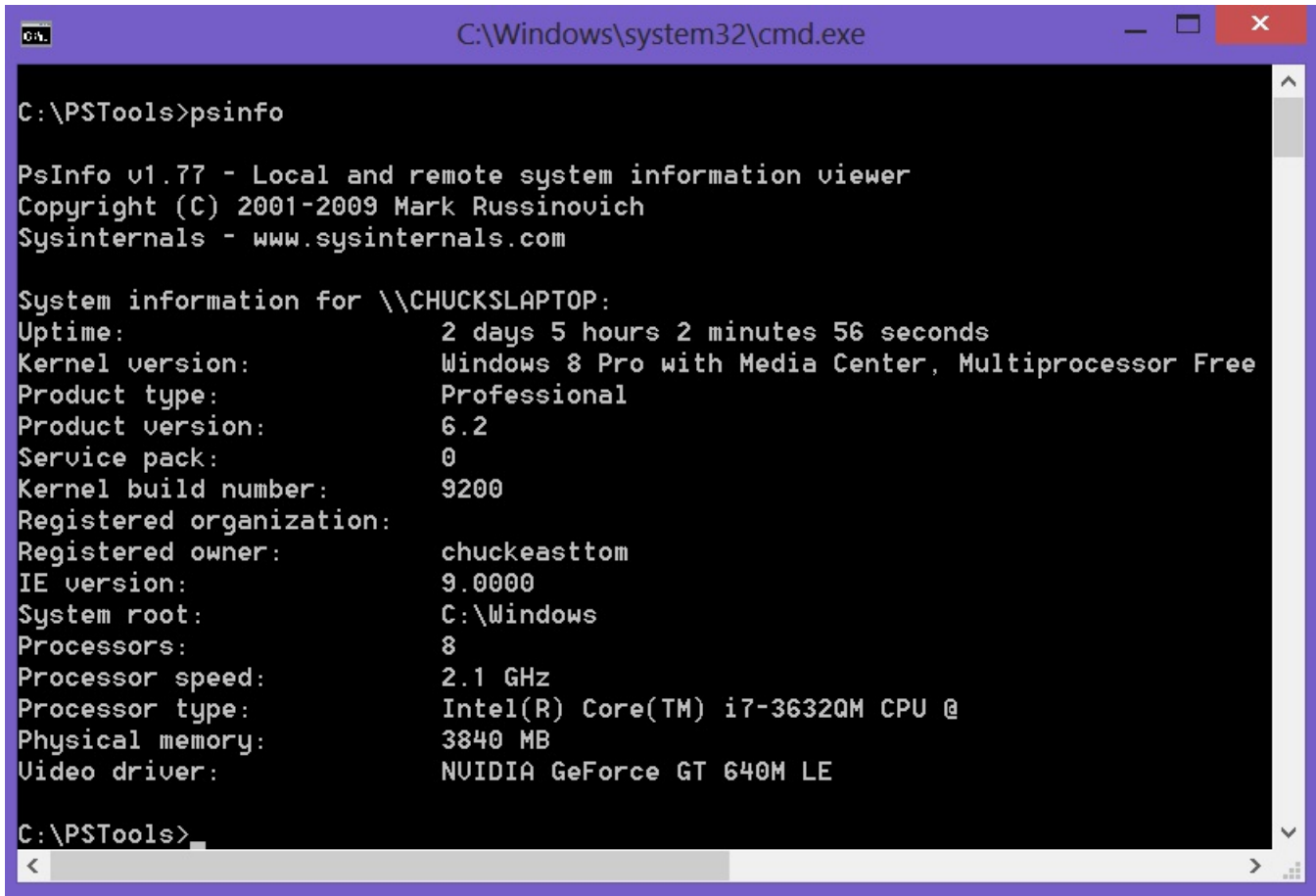
```
C:\Windows\system32\cmd.exe
Copyright (C) 2000-2012 Mark Russinovich
Sysinternals - www.sysinternals.com

Process information for CHUCKSLAPTOP:

Name                Pid Pri Thd Hnd Priv CPU Time Elapsed Time
Idle                 0  0  8   0   0 333:07:39.906 0:00:00.000
System               4  8 150 1070 144  4:10:51.781 53:01:22.315
smss                 340 11  2  36  288 0:00:00.125 53:01:22.190
csrss                512 13 10 586 2544 0:00:04.531 53:01:12.477
csrss                652 13 13 539 4032 0:02:58.328 53:01:10.491
wininit              660 13  2  71  928 0:00:00.109 53:01:10.476
winlogon             716 13  3 141 1436 0:00:00.703 53:01:09.366
services             760  9 14 427 8152 0:00:13.718 53:01:08.585
lsass                768  9  8 1276 10696 0:01:54.015 53:01:08.319
svchost              872  8  7 398 4280 0:00:02.046 53:01:03.007
nvusvc               908  8  4 140 2044 0:00:00.140 53:01:02.679
svchost              948  8  9 785 8836 0:00:18.921 53:01:02.163
svchost             1000  8 26 878 22080 0:00:24.468 53:01:01.694
dwm                  356 13  5 254 28940 0:51:45.265 53:01:01.413
svchost              384  8 43 1187 140972 0:37:10.203 53:01:01.288
svchost              524  8 56 2192 269672 0:25:43.718 53:01:01.288
svchost              500  8 27 857 23852 0:00:23.781 53:01:00.678
nvxdsync            1076  8  9 303 6288 0:00:01.703 53:00:59.491
nvusvc              1096  8  4 191 4960 0:00:00.406 53:00:59.428
svchost             1192  8 18 649 15920 0:00:56.390 53:00:57.866
spoolsv              1572  8 25 541 10804 0:00:06.218 53:00:51.902
svchost             1600  8 21 506 29540 0:00:57.046 53:00:51.746
AppleMobileDeviceService 1740  8  9 228 2940 0:00:32.421 53:00:49.542
mDNSResponder        1972  8  4 125 1816 0:00:01.968 53:00:41.542
EvtEng               2032  8 20 274 5704 0:00:02.187 53:00:41.292
dasHost              1048  8 11 374 5928 0:00:01.781 53:00:41.073
HeciServer            848  8  3 100 1296 0:00:00.015 53:00:40.636
Jhi_service          1556  8  3  75 1176 0:00:00.015 53:00:40.464
mfEvtps              1892  8  4 176 1448 0:00:00.031 53:00:39.589
MsDtsSrvr            2076  8 14 358 97824 0:00:01.281 53:00:38.901
```

Used with permission from Microsoft

PsInfo



A screenshot of a Windows command prompt window titled "C:\Windows\system32\cmd.exe". The window shows the execution of the "psinfo" command from the "C:\PSTools" directory. The output displays system information for the machine named "\\CHUCKSLAPTOP".

```
C:\PSTools>psinfo

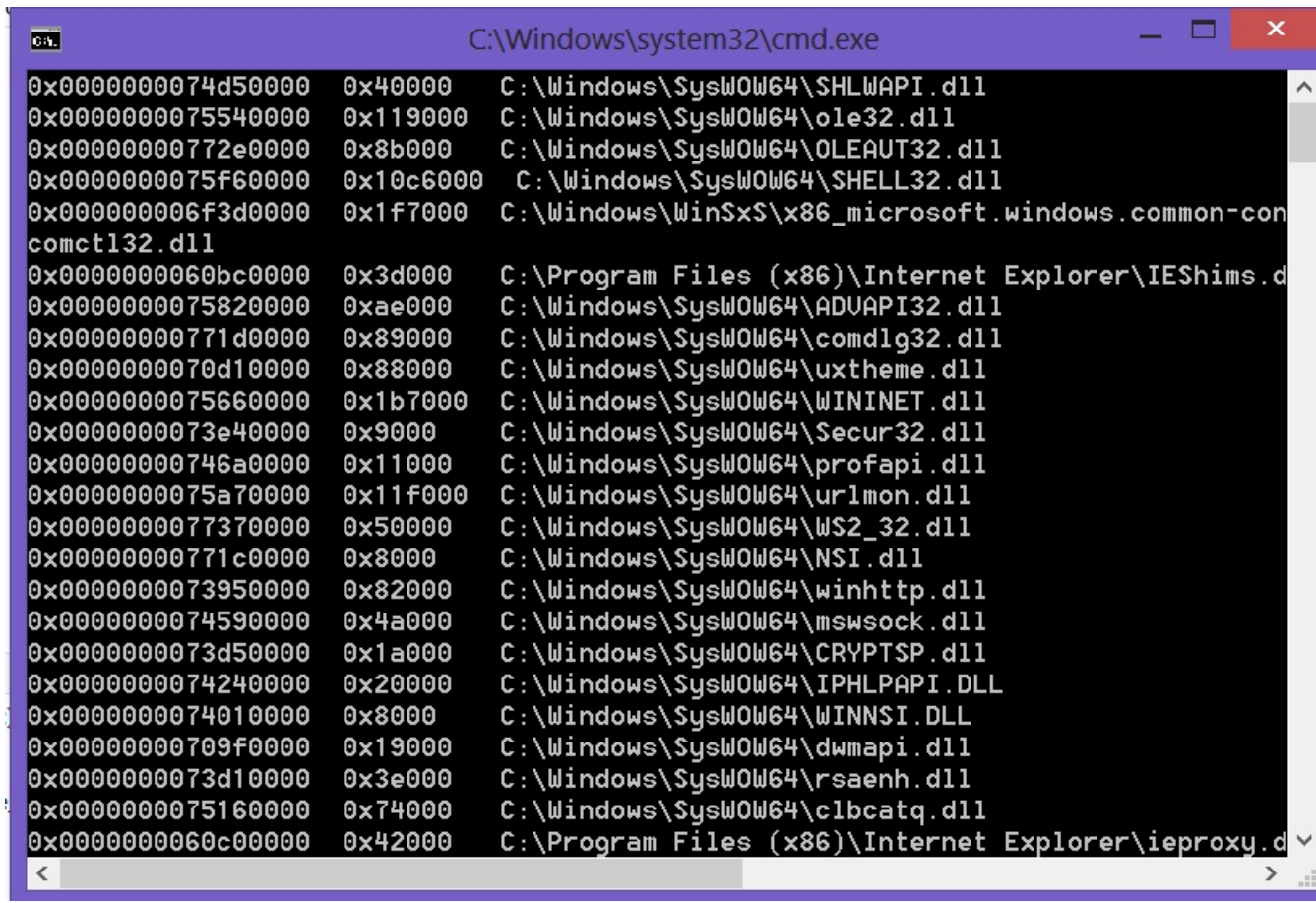
PsInfo v1.77 - Local and remote system information viewer
Copyright (C) 2001-2009 Mark Russinovich
Sysinternals - www.sysinternals.com

System information for \\CHUCKSLAPTOP:
Uptime:                2 days 5 hours 2 minutes 56 seconds
Kernel version:        Windows 8 Pro with Media Center, Multiprocessor Free
Product type:          Professional
Product version:       6.2
Service pack:          0
Kernel build number:   9200
Registered organization:
Registered owner:      chuckeasttom
IE version:            9.0000
System root:           C:\Windows
Processors:            8
Processor speed:       2.1 GHz
Processor type:        Intel(R) Core(TM) i7-3632QM CPU @
Physical memory:       3840 MB
Video driver:          NVIDIA GeForce GT 640M LE

C:\PSTools>
```

Used with permission from Microsoft

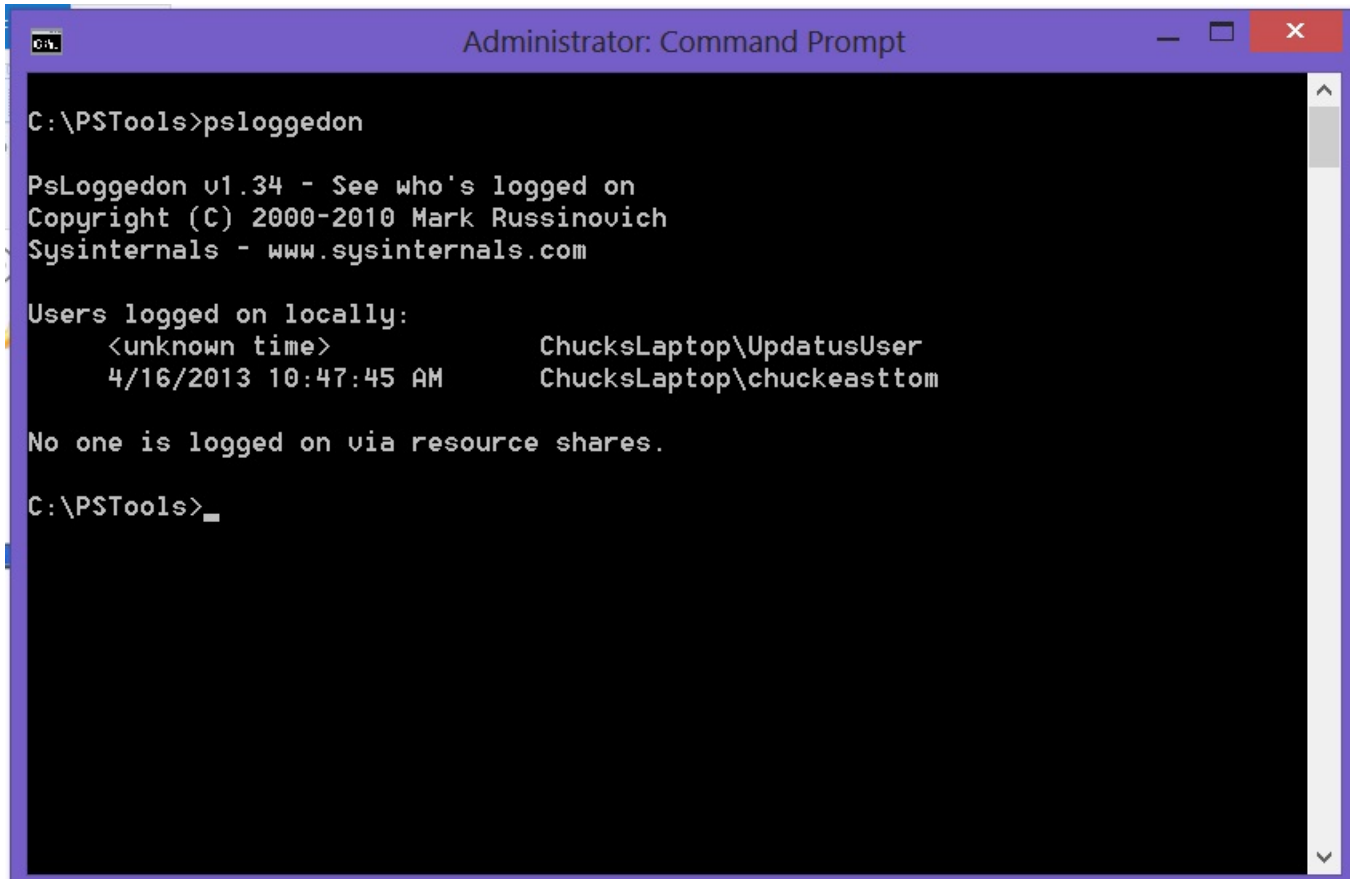
ListDLLs



```
C:\Windows\system32\cmd.exe
0x0000000074d50000 0x40000 C:\Windows\SysWOW64\SHLWAPI.dll
0x0000000075540000 0x119000 C:\Windows\SysWOW64\ole32.dll
0x00000000772e0000 0x8b000 C:\Windows\SysWOW64\OLEAUT32.dll
0x0000000075f60000 0x10c6000 C:\Windows\SysWOW64\SHELL32.dll
0x000000006f3d0000 0x1f7000 C:\Windows\WinSxS\x86_microsoft.windows.common-con
comctl32.dll
0x0000000060bc0000 0x3d000 C:\Program Files (x86)\Internet Explorer\IEShims.d
0x0000000075820000 0xae000 C:\Windows\SysWOW64\ADVAPI32.dll
0x00000000771d0000 0x89000 C:\Windows\SysWOW64\comdlg32.dll
0x0000000070d10000 0x88000 C:\Windows\SysWOW64\uxtheme.dll
0x0000000075660000 0x1b7000 C:\Windows\SysWOW64\WININET.dll
0x0000000073e40000 0x9000 C:\Windows\SysWOW64\Secur32.dll
0x00000000746a0000 0x11000 C:\Windows\SysWOW64\profapi.dll
0x0000000075a70000 0x11f000 C:\Windows\SysWOW64?urlmon.dll
0x0000000077370000 0x50000 C:\Windows\SysWOW64\WS2_32.dll
0x00000000771c0000 0x8000 C:\Windows\SysWOW64\NSI.dll
0x0000000073950000 0x82000 C:\Windows\SysWOW64\winhttp.dll
0x0000000074590000 0x4a000 C:\Windows\SysWOW64\mswsock.dll
0x0000000073d50000 0x1a000 C:\Windows\SysWOW64\CRYPTSP.dll
0x0000000074240000 0x20000 C:\Windows\SysWOW64\IPHLPAPI.DLL
0x0000000074010000 0x8000 C:\Windows\SysWOW64\WINNSI.DLL
0x00000000709f0000 0x19000 C:\Windows\SysWOW64\dwmapi.dll
0x0000000073d10000 0x3e000 C:\Windows\SysWOW64\rsaenh.dll
0x0000000075160000 0x74000 C:\Windows\SysWOW64\clbcatq.dll
0x0000000060c00000 0x42000 C:\Program Files (x86)\Internet Explorer\ieproxy.d
```

Used with permission from Microsoft

PsLoggedOn



```
C:\PSTools>psloggedon

PsLoggedon v1.34 - See who's logged on
Copyright (C) 2000-2010 Mark Russinovich
Sysinternals - www.sysinternals.com

Users logged on locally:
    <unknown time>          ChucksLaptop\UpdatusUser
    4/16/2013 10:47:45 AM    ChucksLaptop\chuckeasttom

No one is logged on via resource shares.

C:\PSTools>
```

Used with permission from Microsoft

Netstat

```
C:\>netstat -an

Active Connections

Proto Local Address          Foreign Address         State
TCP   0.0.0.0:80              0.0.0.0:0               LISTENING
TCP   0.0.0.0:135            0.0.0.0:0               LISTENING
TCP   0.0.0.0:443            0.0.0.0:0               LISTENING
TCP   0.0.0.0:445            0.0.0.0:0               LISTENING
TCP   0.0.0.0:554            0.0.0.0:0               LISTENING
TCP   0.0.0.0:1025           0.0.0.0:0               LISTENING
TCP   0.0.0.0:1026           0.0.0.0:0               LISTENING
TCP   0.0.0.0:1027           0.0.0.0:0               LISTENING
TCP   0.0.0.0:1028           0.0.0.0:0               LISTENING
TCP   0.0.0.0:1032           0.0.0.0:0               LISTENING
TCP   0.0.0.0:1074           0.0.0.0:0               LISTENING
TCP   0.0.0.0:1583           0.0.0.0:0               LISTENING
TCP   0.0.0.0:1790           0.0.0.0:0               LISTENING
TCP   0.0.0.0:2382           0.0.0.0:0               LISTENING
TCP   0.0.0.0:2603           0.0.0.0:0               LISTENING
TCP   0.0.0.0:2869           0.0.0.0:0               LISTENING
TCP   0.0.0.0:3351           0.0.0.0:0               LISTENING
TCP   0.0.0.0:5357           0.0.0.0:0               LISTENING
TCP   0.0.0.0:6002           0.0.0.0:0               LISTENING
```

Used with permission from Microsoft

Other Live Forensics Tools

FPort

- View all open TCP and UDP ports
- Map ports to specific processes

Userdump

- Extract memory dumps of running processes

PTFinder

- Enumerates processes and threads in a memory dump

Windows Swap File

- A special place on the hard disk where items from memory can be temporarily stored for fast retrieval
- Used to end in a .swp extension; since Windows XP, called *pagefile.sys*
- Typically found in Windows root directory
- Often referred to as virtual memory

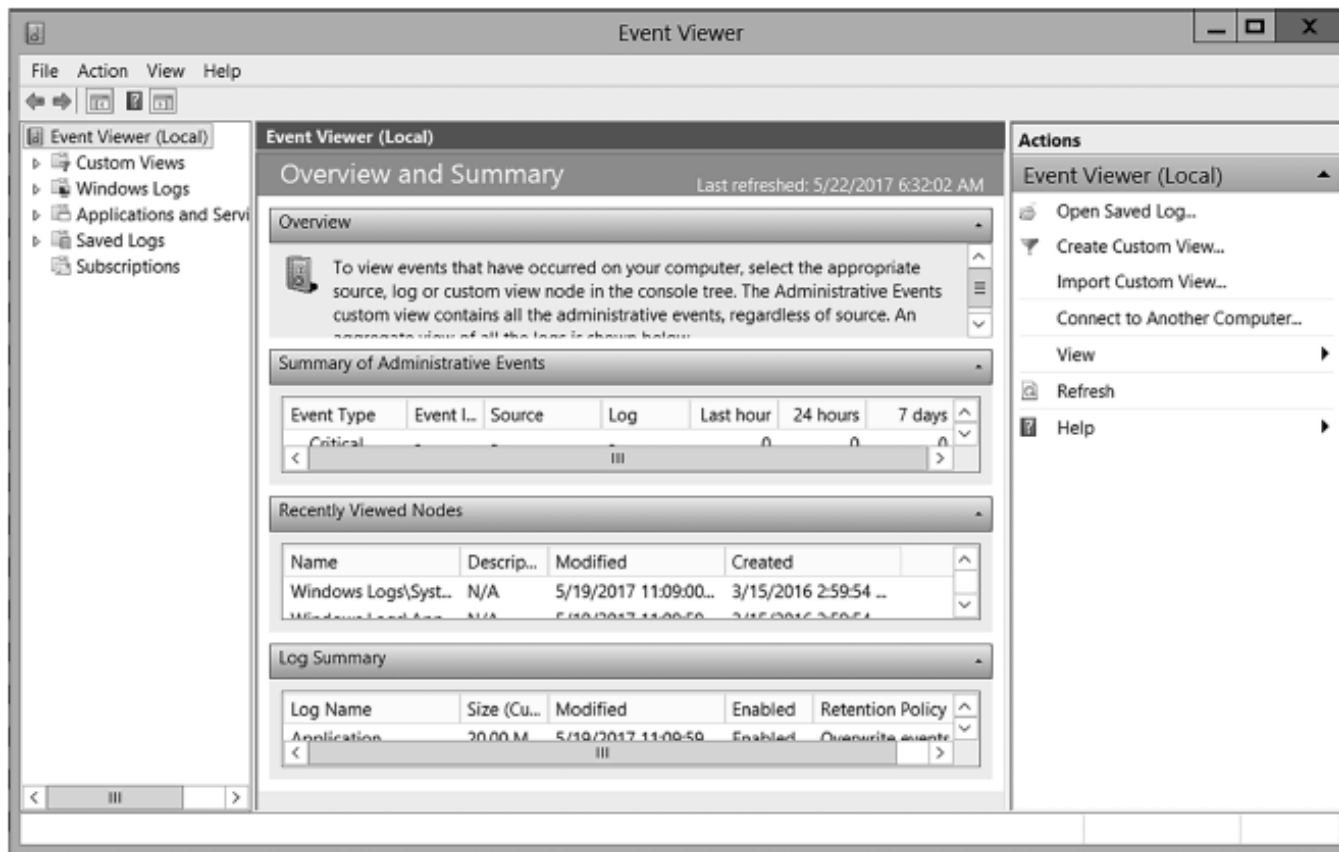
Windows Log Files

- Files that contain information about events and other activities that occur in Windows
- Event Viewer used to view log files

Windows Log Files



Event Viewer



Used with permission from Microsoft

Windows Directories/Folders

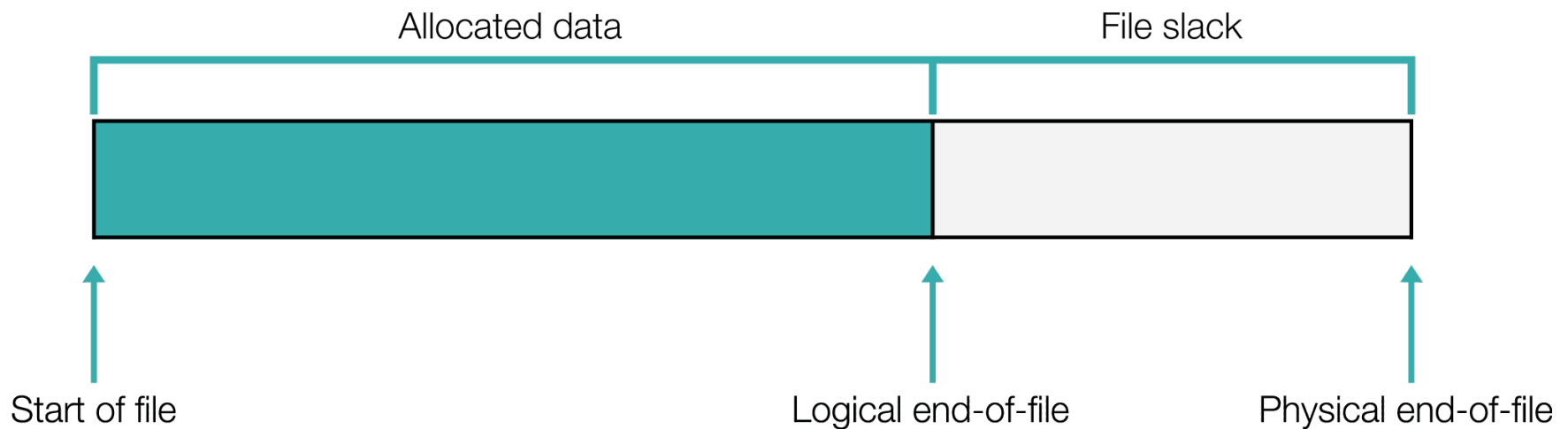
- C:\Windows documents and settings
- C:\users
- C:\Program Files
- C:\Program Files (x86)
- C:\Users\username\Documents

UserAssist

UserAssist 2.5.0.0									
Commands Help									
Key	Index	Name	Unkno...	Sessi...	Counter	Last	Last UTC	Focus count...	Fc ^
{CEBF...	0	UEME_CTLCUACount ctor			0			0	
{CEBF...	1	Microsoft.Windows.Explorer			0			533	
{CEBF...	2	UEME_CTLSESSION							
{CEBF...	3	Microsoft.Windows.ControlPanel			0			24	
{CEBF...	4	Microsoft.InternetExplorer.Default			43	4/19/2013 1:42:33 PM	4/19/2013 6:42:33 PM	860	
{CEBF...	5	C:\Users\chuckeasttom\AppData\Local\Microsoft\Windows\Te...			0	1/30/2013 5:00:07 PM	1/30/2013 11:00:07 PM	0	
{CEBF...	6	C:\Users\chuckeasttom\AppData\Local\Amazon\Kindle\applica...			2	4/19/2013 12:10:08 PM	4/19/2013 5:10:08 PM	16	
{CEBF...	7	D:\SETUP.EXE			8	4/16/2013 10:13:54 AM	4/16/2013 3:13:54 PM	0	
{CEBF...	8	{1AC14E77-02E7-4E5D-B744-2EB1AE5198B7}\rundll32.exe			0	4/14/2013 9:10:46 PM	4/15/2013 2:10:46 AM	0	
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{CEBF...	10	D:\autorun.exe			0	4/14/2013 9:06:15 PM	4/15/2013 2:06:15 AM	0	
{CEBF...	11	{7C5A40EF-A0FB-4BFC-874A-C0F2E0B9FA8E}\Common Files\L...			0			0	
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{CEBF...	15	{7C5A40EF-A0FB-4BFC-874A-C0F2E0B9FA8E}\Microsoft Office...			8	4/19/2013 1:41:33 PM	4/19/2013 6:41:33 PM	1006	
{CEBF...	16	microsoft.windows.photos_8wekyb3d8bbwe\Microsoft.WindowsL...			0			0	
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{CEBF...	20	{7C5A40EF-A0FB-4BFC-874A-C0F2E0B9FA8E}\Microsoft Office...			56	4/19/2013 2:53:32 PM	4/19/2013 7:53:32 PM	528	
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{CEBF...	23	Microsoft.AutoGenerated.{8ABD94FB-E7D6-8A6-A997-C918E...			0	2/2/2013 2:05:29 PM	2/2/2013 8:05:29 PM	0	
{CEBF...	24	{1AC14E77-02E7-4E5D-B744-2EB1AE5198B7}\msconfig.exe			0	3/26/2013 10:41:01 AM	3/26/2013 3:41:01 PM	0	
{CEBF...	25	Apple.iTunes			0			85	
{CEBF...	26	Microsoft.DSUI.Device.{0C9DEDE0-3428-5A1C-8029-FB940833...			0			0	
{CEBF...	27	Microsoft.DSUI.Device.{A343F791-6C8A-11E2-BE75-C8F7331D...			0			0	
{CEBF...	28	{7C5A40EF-A0FB-4BFC-874A-C0F2E0B9FA8E}\iTunes\iTunes....			0	3/18/2013 1:24:31 PM	3/18/2013 6:24:31 PM	0	
{CEBF...	29	{1AC14E77-02E7-4E5D-B744-2EB1AE5198B7}\FileHistory.exe			0			0	
{CEBF...	30	{1AC14E77-02E7-4E5D-B744-2EB1AE5198B7}\DevicePairingW...			0	2/3/2013 8:34:18 PM	2/4/2013 2:34:18 AM	0	
{CEBF...	31	{1AC14E77-02E7-4E5D-B744-2EB1AE5198B7}\mblctr.exe			0			0	

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Unallocated/Slack Space



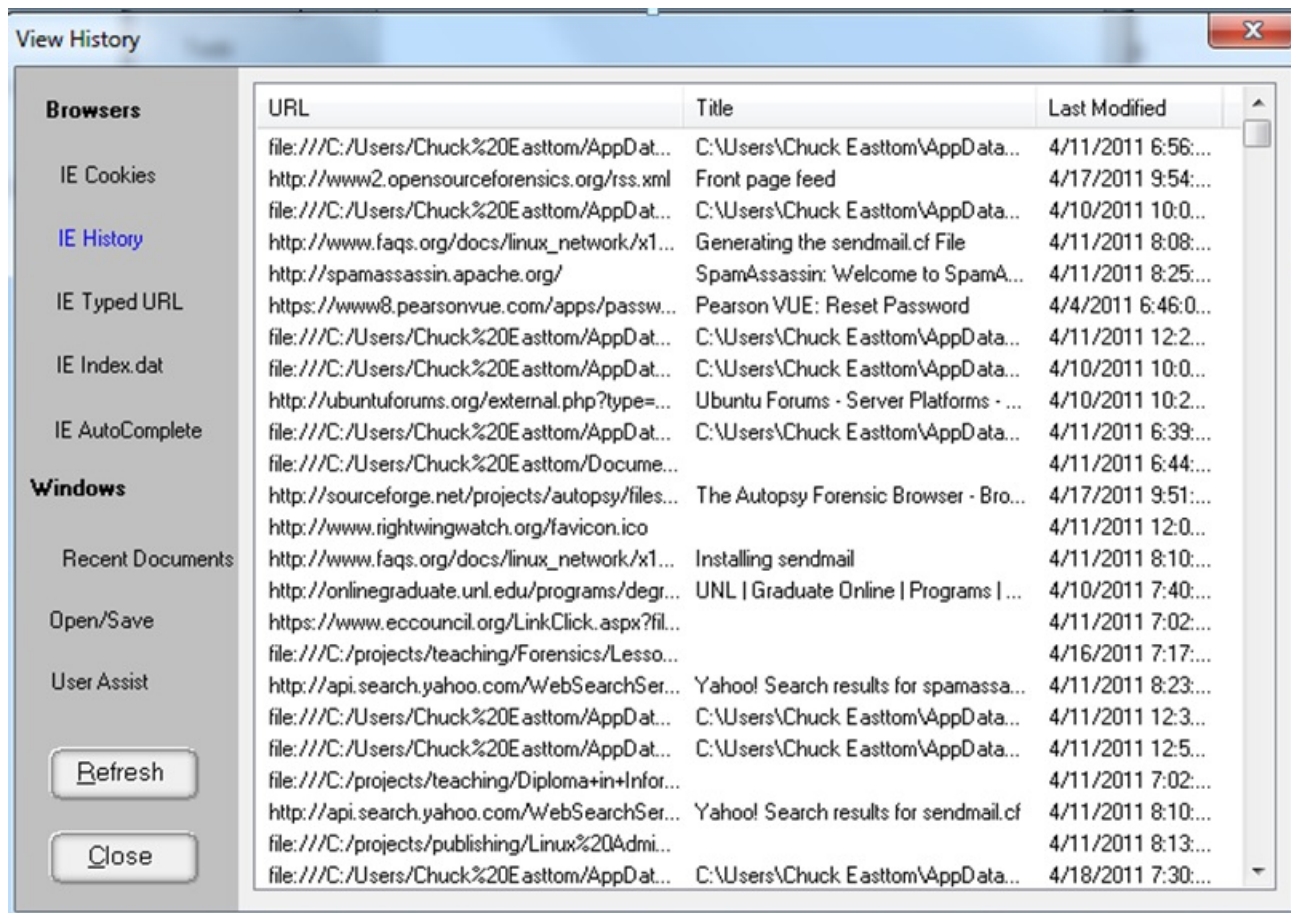
Alternate Data Streams (ADS)

- A method of attaching one file to another file, using the NTFS file system
- A feature of NTFS that contains metadata for locating a specific file by some criterion, like title
- What are the risks associated with ADS?

Index.dat

- Used by Microsoft Internet Explorer
- Stores:
 - Web addresses
 - Search queries
 - Recently opened files

Window Washer



Courtesy of Eusing Software

Windows Files and Permissions

- When copying and pasting on the same partition, files and folders **inherit the rights of the folder** they are being copied to.
- When cutting and pasting (moving), files and folders **retain the original permissions** if they are on the same partition.

Windows Files and Permissions (Cont.)

- MAC refers to three critical properties:
 - File modified
 - File accessed
 - File created

Windows Registry

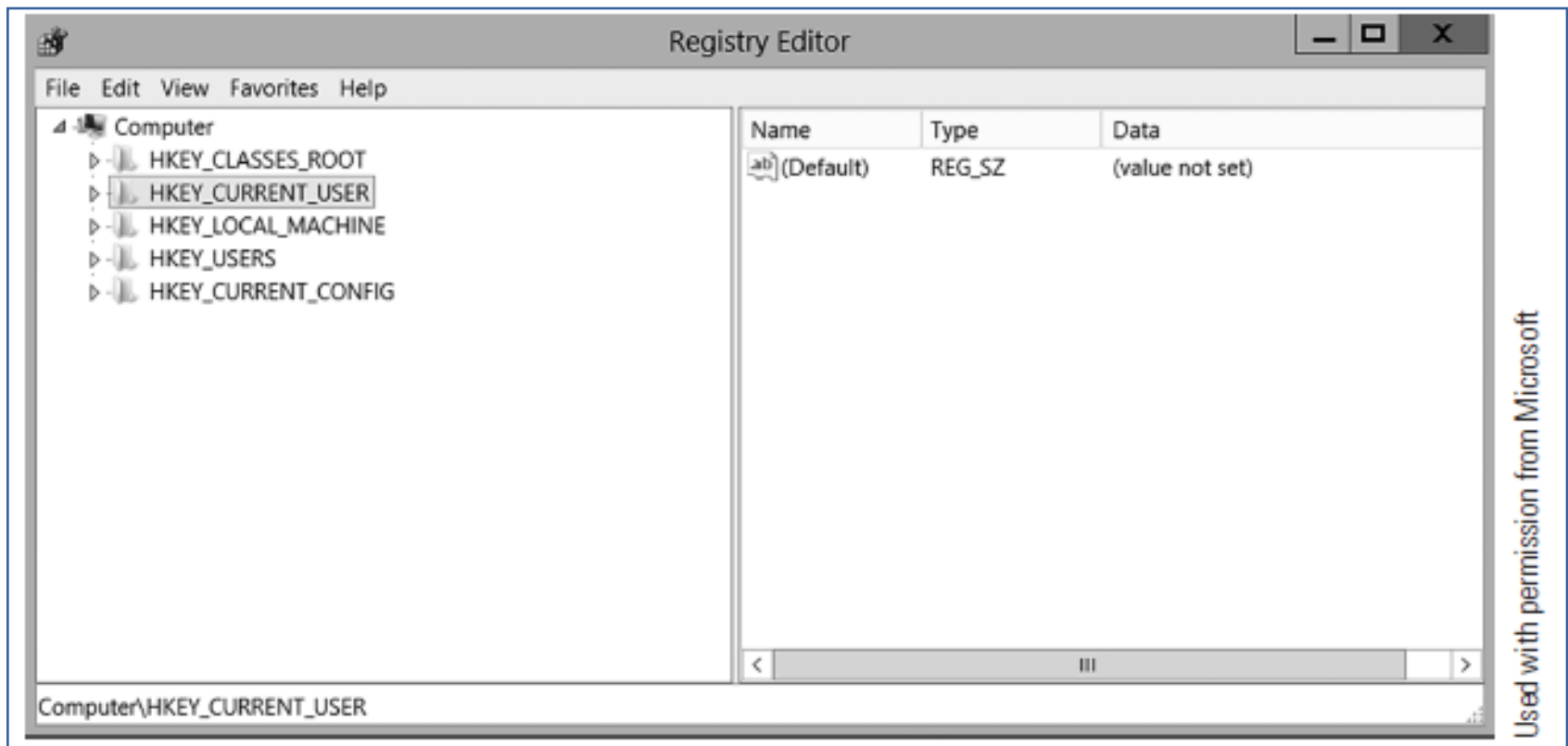
Computer hardware configuration

Multiple users and preferences

Program shortcuts and properties sheets

Remote administration through network

Windows Registry



Windows Registry Hives

- HKEY_CLASSES_ROOT (HKCR)
- HKEY_CURRENT_USER (HKCU)
- HKEY_LOCAL_MACHINE (HKLM)
- HKEY_USERS (HKU)
- HKEY_CURRENT_CONFIG (HCU)

Importance of Examining Windows Registry

USB

Wireless
network

Word
documents

Malware

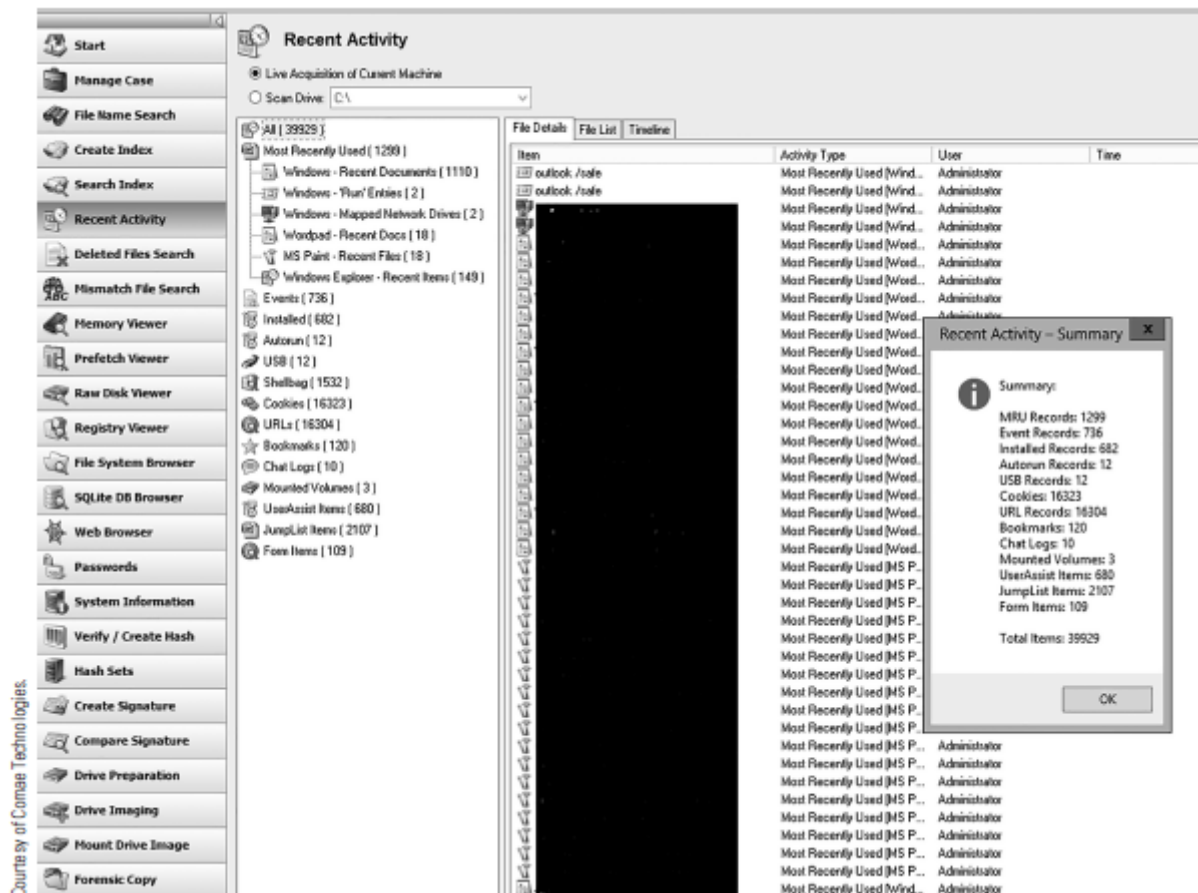
Uninstalled
software

Passwords

ShellBag

Prefetch

Prefetch



Courtesy of Comae Technologies

Volume Shadow Copy

- Keeps a record or copy of state changes
- Stores them in blocks of data that are compared daily
- Changed blocks are copied to Volume Shadow
- Volume Shadow Copy service runs once per day

Memory Forensics

1. Capture the memory from a live machine.

Can use:

- Dump-it, RAM Capturer from Belkasoft, OSForensics, other tools

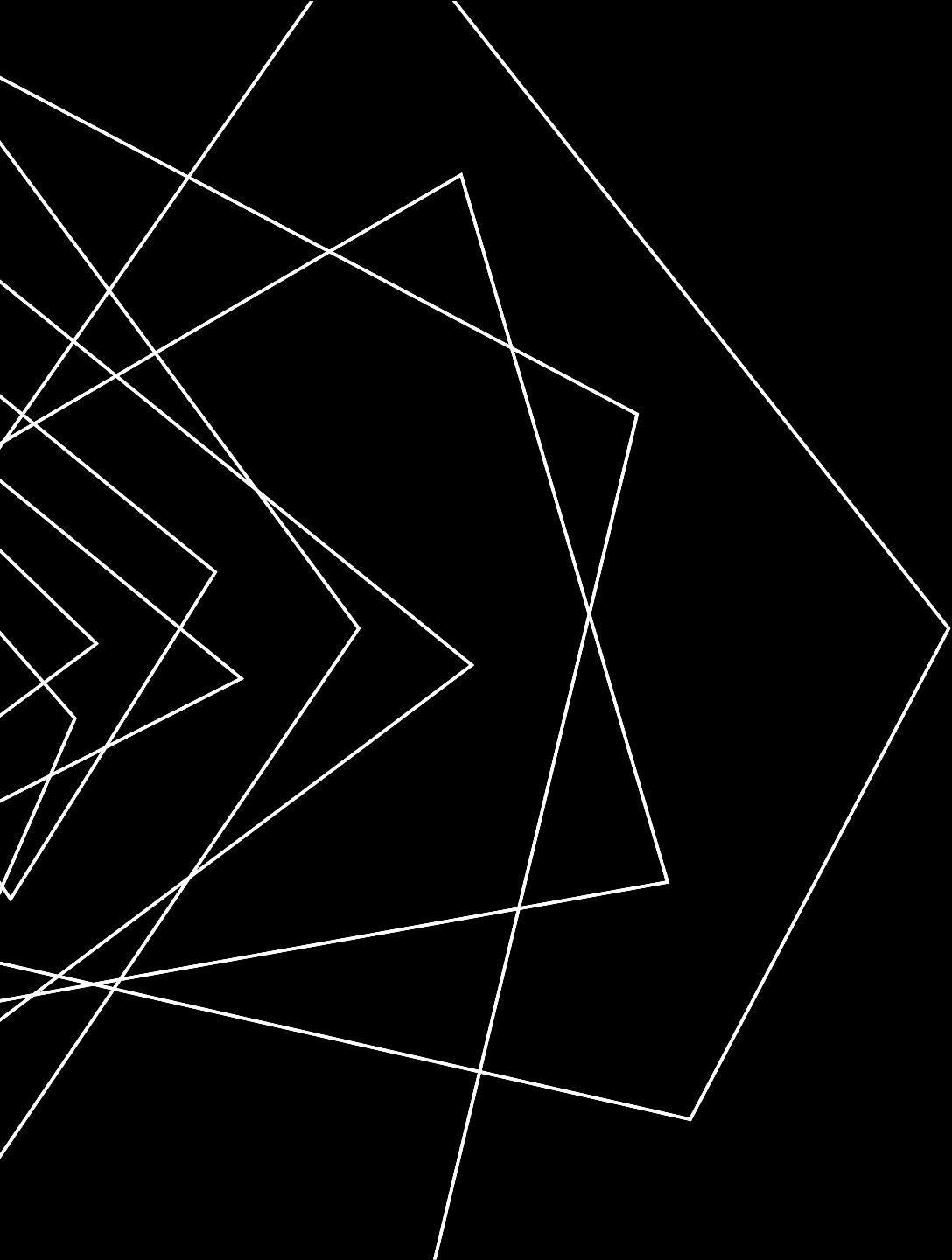
2. Analyze the captured memory.

Can use:

- Volatility, Pslist, Pstree, Psscan, Svcscan, other tools

Summary

- Windows details
- Evidence in volatile data
- Windows swap file
- Windows logs and directories
- Windows Registry



THANK YOU

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